

# GD&T Symbols Chart

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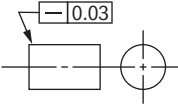
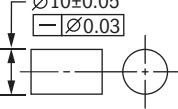
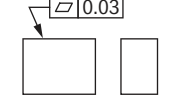
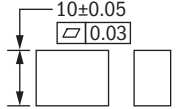
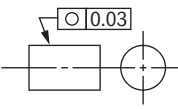
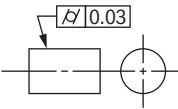
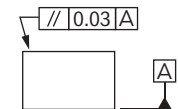
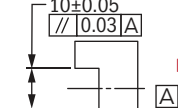
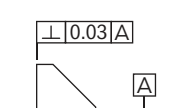
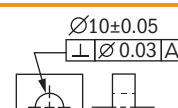
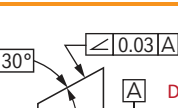
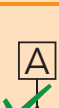
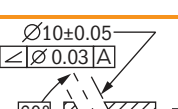
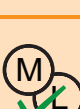
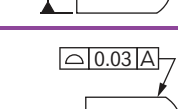
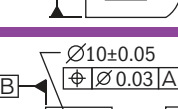
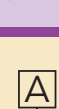
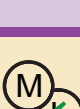
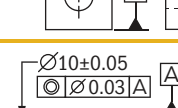
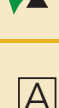
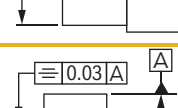
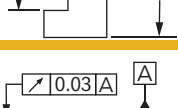
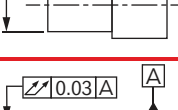
FORM

ORIENTATION

PROFILE

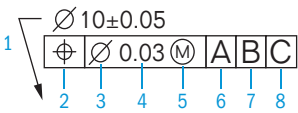
LOCATION

RUNOUT

| SYMBOL                                                                                                                                      | THIS ON THE DRAWING                                                                 | MEANS THIS TOLERANCE                                                                                                   | DATUMS & MODIFIERS                                                                                                                                                         |
|---------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div></div> <b>Straightness</b><br>(Surface)               |    | CONTROLS CROSS-SECTIONAL ELEMENTS TO (2) PARALLEL LINES 0.03 APART                                                     |       |
| <div></div> <b>Straightness</b><br>(Feature of Size)       |    | CONTROLS DERIVED MEDIAN LINE TO Ø0.03 CYLINDRICAL ZONE                                                                 |       |
| <div></div> <b>Flatness</b><br>(Surface)                   |    | CONTROLS SURFACE ELEMENTS TO (2) PARALLEL PLANES 0.03 APART                                                            |       |
| <div></div> <b>Flatness</b><br>(Feature of Size)           |    | CONTROLS DERIVED MEDIAN PLANE TO (2) PARALLEL PLANES 0.03 APART                                                        |       |
| <div></div> <b>Circularity</b>                             |    | CONTROLS CROSS-SECTIONAL ELEMENTS TO (2) CONCENTRIC CIRCLES RADIALLY 0.03 APART                                        |       |
| <div></div> <b>Cylindricity</b>                          |  | CONTROLS SURFACE ELEMENTS TO (2) CONCENTRIC CYLINDERS RADIALLY 0.03 APART                                              |   |
| <div></div> <b>Parallelism</b><br>(Surface)              |  | CONTROLS SURFACE ELEMENTS TO (2) PARALLEL PLANES 0.03 APART<br>ORIENTED TO DATUM(S)                                    |   |
| <div></div> <b>Parallelism</b><br>(Feature of Size)      |  | CONTROLS MIDPLANE (OR AXIS) TO (2) PARALLEL PLANES 0.03 APART<br>ORIENTED TO DATUM(S)                                  |   |
| <div></div> <b>Perpendicularity</b><br>(Surface)         |  | CONTROLS SURFACE ELEMENTS TO (2) PARALLEL PLANES 0.03 APART<br>ORIENTED TO DATUM(S)                                    |   |
| <div></div> <b>Perpendicularity</b><br>(Feature of Size) |  | CONTROLS AXIS (OR MIDPLANE) TO Ø0.03 CYLINDRICAL ZONE<br>ORIENTED TO DATUM(S)                                          |   |
| <div></div> <b>Angularity</b><br>(Surface)               |  | CONTROLS SURFACE ELEMENTS TO (2) PARALLEL PLANES 0.03 APART<br>ORIENTED TO DATUM(S)                                    |   |
| <div></div> <b>Angularity</b><br>(Feature of Size)       |  | CONTROLS AXIS (OR MIDPLANE) TO Ø0.03 CYLINDRICAL ZONE<br>ORIENTED TO DATUM(S)                                          |   |
| <div></div> <b>Profile of Line</b>                       |  | CONTROLS CROSS-SECTIONAL ELEMENTS TO UNIFORM 2D BOUNDARY ABOUT TRUE PROFILE 0.03 APART<br>LOCATED/ORIENTED TO DATUM(S) |   |
| <div></div> <b>Surface Profile</b>                       |  | CONTROLS SURFACE ELEMENTS TO UNIFORM 3D BOUNDARY ABOUT TRUE PROFILE 0.03 APART<br>LOCATED/ORIENTED TO DATUM(S)         |   |
| <div></div> <b>Position</b>                              |  | CONTROLS AXIS (OR MIDPLANE) TO Ø0.03 CYLINDRICAL ZONE<br>LOCATED/ORIENTED TO DATUM(S)                                  |   |
| <div></div> <b>Concentricity</b>                         |  | CONTROLS DERIVED MEDIAN POINTS TO Ø0.03 CYLINDRICAL ZONE<br>LOCATED/ORIENTED TO DATUM                                  |   |
| <div></div> <b>Symmetry</b>                              |  | CONTROLS DERIVED MEDIAN POINTS TO (2) PARALLEL PLANES 0.03 APART<br>LOCATED/ORIENTED TO DATUM                          |   |
| <div></div> <b>Runout</b>                                |  | CONTROLS CROSS-SECTIONAL ELEMENTS TO (2) COAXIAL CIRCLES RADIALLY 0.03 APART<br>LOCATED/ORIENTED TO DATUM              |   |
| <div></div> <b>Total Runout</b>                          |  | CONTROLS SURFACE ELEMENTS TO (2) COAXIAL CYLINDERS RADIALLY 0.03 APART<br>LOCATED/ORIENTED TO DATUM                    |   |

Contact Us: 1-800-495-0991 or [info@gdandtbasics.com](mailto:info@gdandtbasics.com)

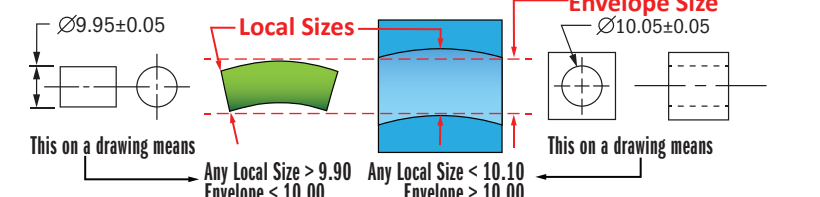
### Feature Control Frame Components



- Leader Arrow
- Geometric Control Symbol
- Tolerance Zone Shape
- Geometric Tolerance Size
- Modifiers
- Primary Datum Feature Ref.
- Secondary Datum Feature Ref.
- Tertiary Datum Feature Ref.

#### Rule # 1 of GD&T: Envelope Principle

Rule #1 states that the surface of a regular feature of size can not extend beyond the envelope of perfect form at Maximum Material Condition (MMC)



This on a drawing means: Any Local Size > 9.90 Envelope < 10.00

#### Rule # 2: Implied Regardless of Feature Size

Regardless of Feature Size is always implied on geometric tolerances unless otherwise specified by Least Material Condition or Maximum Material Condition modifiers.

#### RFS - Regardless of Feature Size

Geometric tolerances need to be within tolerance regardless of the size of the feature. This is the default condition on all GD&T unless otherwise specified - See Rule # 2 above.

#### Maximum Material Condition Modifier (MMC)

Indicates this specific geometric tolerance is to be assessed at maximum material condition. Bonus geometric tolerance is allowed as the size of the feature deviates away from maximum material condition.

Bonus Tolerance = Actual size deviation from the Max Material Condition

#### Least Material Condition Modifier (LMC)

Indicates this specific geometric tolerance is to be assessed at least material condition. Bonus geometric tolerance is allowed as the size of the feature deviates away from least material condition.

Bonus Tolerance = Actual size deviation from the Least Material Condition

#### Virtual Condition

A static boundary created by the effects of a feature of size's specified MMC or LMC and the geometric tolerance for that material condition.

For internal features like a hole:  
Virtual Condition = MMC - Geometric tolerance

For external features like a pin:  
Virtual Condition = MMC + Geometric tolerance

#### Datum Feature Symbol

Indicates a tangible surface or feature of size that is as referenced by a feature control frame.

#### Datum Target

Reference to datum points, lines or areas needed to create a theoretical datum or datum reference frame for part measurement.

#### Projected Tolerance

The tolerance zone is extended out from the surface where the feature begins to a specified height. Used commonly with position to represent the virtual condition of a pin stud

#### Free State

Used to indicate that a specific dimension and its associated tolerance are to be assessed in an unrestrained state. By default, all dimensions are assumed to be evaluated in a "Free State".

#### Independency

When indicated, Rule #1 is overridden and the part's geometric tolerance is no longer restricted by the limits of size.

#### Continuous Feature

Used to indicate that a group of two or more features are to be considered as one continuous feature. Rule #1 and any geometric controls will be applied to the considered features as if they are one.

#### Tangent Plane

A theoretical tangent plane is established on the surface of a part based on the surface high-points. When called, only the virtual tangent plane needs to be within the tolerance zone, instead of all the surface itself.

#### Basic Dimensions

Dimensions that are listed in a rectangular box that define theoretically exact dimensions. When defining part geometry, GD&T controls will dictate the tolerance for these dimensions as they are untoleranced by themselves.

#### Between Symbol (For Profile Only)

Specifies the exact limits of the surface or line (usually encompassing multiple features) that the profile tolerance controls. This is indicated by labeling two extremities in a single view.

#### All Around Symbol (For Profile Only)

Indicates that the geometric tolerance applies to every surface around the part in the specified view.

#### All Over Symbol (For Profile Only)

Indicates that the geometric tolerance applies to every surface around the entire part in every view (3D)

#### Unequally Disposed Profile Symbol (For Profile Only)

Indicates the tolerance zone for profile is no longer symmetrical to the true profile. The value that PRECEDES the symbol is the TOTAL tolerance. The value FOLLOWING the symbol indicates how much of the total tolerance is 'shifted' in the direction that adds material to the part.

